

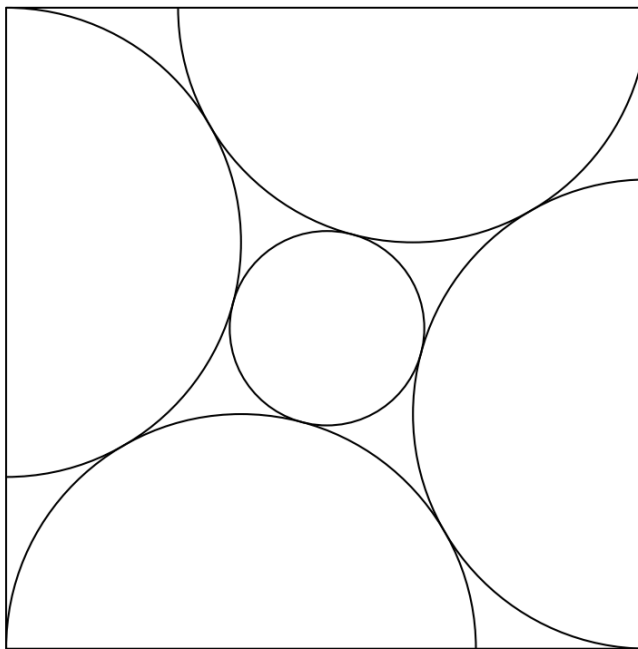
2025 AMC 10B Problem 20 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 20

Problem:

Four congruent semicircles are inscribed in a square of side length 1 so that their diameters are on the sides of the square, one endpoint of each diameter is at a vertex of the square, and adjacent semicircles are tangent to each other. A small circle centered at the center of the square is tangent to each of the four semicircles, as shown below.



The diameter of the small circle can be written as $(\sqrt{a} + b)(\sqrt{c} + d)$, where a , b , c , and d are integers. What is $a + b + c + d$?

Answer Choices:

- A. 3
- B. 5
- C. 8
- D. 9
- E. 11

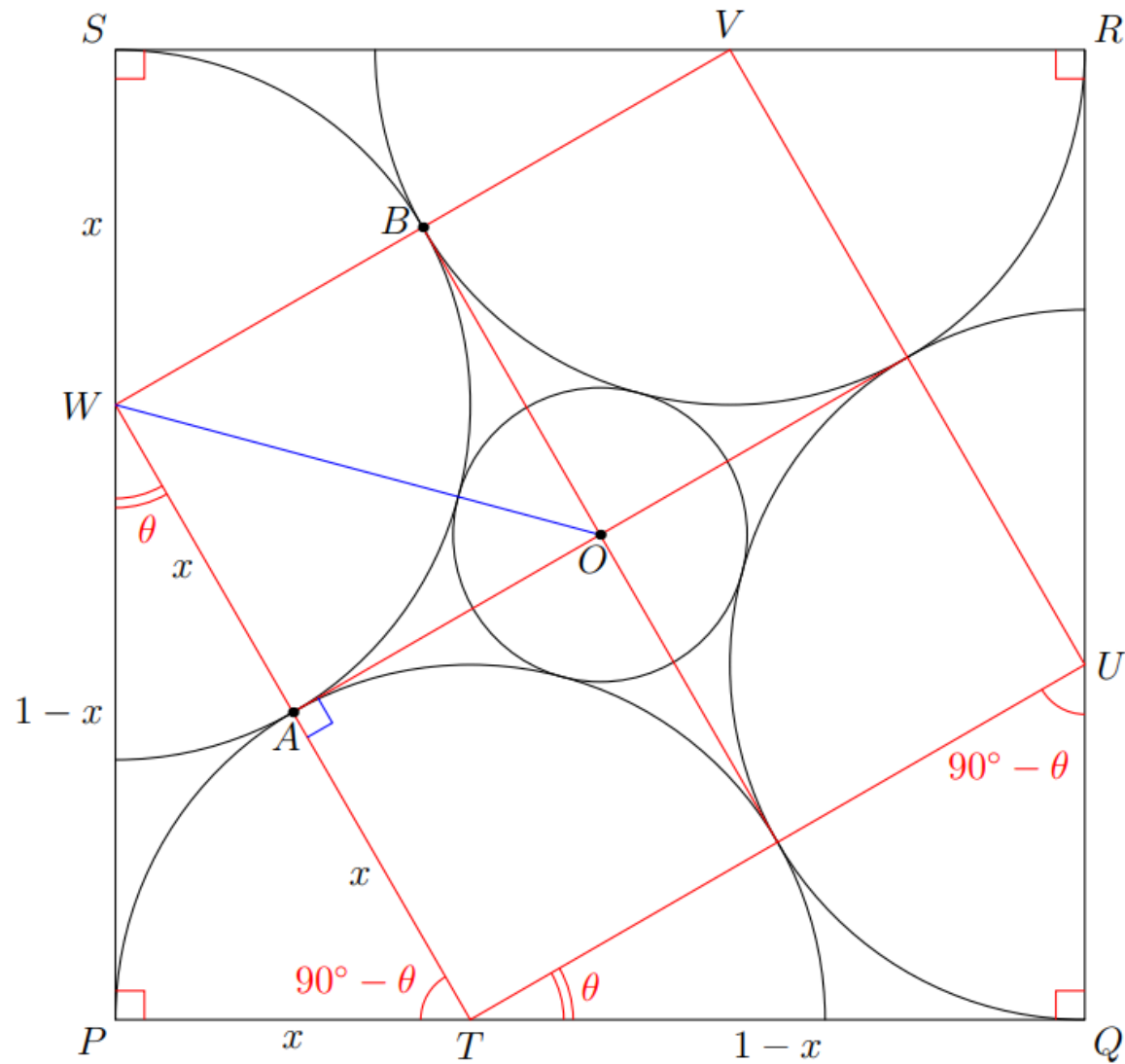
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Solution:

Let the square be $PQRS$, and the semicircles centered at T, U, V, W , as shown below. Let x be the radius of the semicircles.



Note that as the semicircles are tangent, $TW = TA + WA = 2x$ (as TA , AW are radii and T , A , W are collinear). Thus, $\triangle WTP$ is a $30^\circ - 60^\circ - 90^\circ$ triangle, and $WP = 1 - SW = 1 - x$. Thus, we have

$$1 - x = \sqrt{3}x \quad \implies \quad x = \frac{1}{1 + \sqrt{3}} = \frac{\sqrt{3} - 1}{2}$$

Now suppose the small circle has center O . Note that $OW = \sqrt{2} \cdot AW = x\sqrt{2}$ (as $TUVW$ is a square centered at O by symmetry). As the inner circle is tangent to each semicircle,

$$\text{small circle radius} + \text{semi-circle radius} = OW$$

If we let the radius of the small circle be r , we get

$$r + x = x\sqrt{2} \quad \implies \quad r = x \cdot (\sqrt{2} - 1) = \frac{(\sqrt{2} - 1)(\sqrt{3} - 1)}{2}$$

so the diameter is $2r = (\sqrt{2} - 1)(\sqrt{3} - 1)$, which gives $a = 2, b = -1, c = 3, d = -1$. Thus,

$$a + b + c + d = 2 - 1 + 3 - 1 = \boxed{\text{(A) } 3}.$$

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